Portfolio

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Selected projects: CFD for hydrogen injection, topology optimization.

TEAM PROJECT - H2EDGE

2D CFD simulation campaign investigating JICF (jet-in-crossflow) penetration and mixing
Tools: *Ansys Fluent* | Period: *Dec 2025 – Feb 2026*

Goals

- Evaluating the validity of 2D CFD simulation model for an inherently 3D, vortical problem
- Validating jet penetration against empirical correlation
- Investigating feasibility of inert gas (Helium) instead of hydrogen for safer testing approach

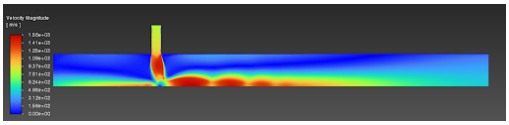


Figure 1: Velocity magnitude - $J \approx 60$

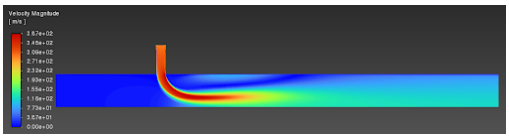


Figure 2: Velocity magnitude - $J \approx 4$

Obtained

- Swept the momentum flux ratio J from ~ 60 to ~ 1.5: at high J the jet chokes and blocks the air upstream; found the sweet spot for ideal penetration around $J \sim 4$ by scaling the problem to overcome the 2D penetration overpredict.
- Inherent mixing deficiency in 2D simulation due to lack of 3D vortical structures implies the need for 3D simulations
- Showed that helium can replace hydrogen for safer testing, provided the density difference is accounted for when matching the momentum flux ratio J.

ACADEMIC PROJECT

Topology optimization of a bracket for aerospace applications
Tools: *Altair Inspire, Bambu Studio* | Period: *Oct 2025 – Dec 2025*

Goals

- Minimizing the mass of a cantilever bracket carrying $F_z = 100$ N and $F_x = 50$ N on the free end, subject to displacement constraints of $U_z < 4$ mm and $U_x < 5$ mm
- Smoothing the geometry, re-verifying it via FEM and adapting it for FDM printing.

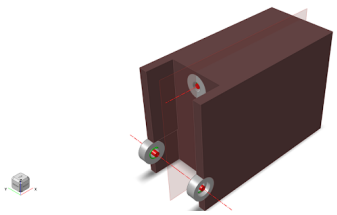


Figure 3: Design before optimization

Obtained

- Mass reduced from $495g$ to $33g$ (-93%) with respect to the pre-optimization design (which was heavily oversized on purpose, in order to give the solver maximum design freedom)
- Linear static FEM check (OptiStruct): max $U_z = 2.81$ mm (limit 4 mm), $U_x = 0.59$mm (limit 5 mm), minimum safety factor 2.6
- Print-oriented variant with overhangs $\leq 45°$: supports reduced from 7.2 g to 1.6 g, print time 1h55 (3h limit) on a Bambu Lab A1

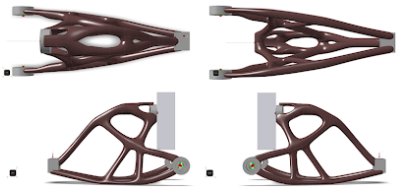


Figure 4: Optimized shape

Multi-agentic workflow for 2D topology optimization software
Tools: *Antigravity agents, Python* | Period: *Aug 2026 – In development*

Goals

- Develop a 2D SIMP solver (Q4 elements) for compliance and linearized buckling via weighted-sum method using a multi-agentic AI-assisted approach.
- Investigate the Pareto-optimal solutions space, i.e. mapping the compliance–buckling trade-off by sweeping the weight α
- Compare and validate its compliance-optimal designs with a commercial solver (Altair Inspire)

Obtained

- Python solver with density filter, Heaviside projection, OC update, adjoint buckling sensitivities and adaptive scaling of the two objective gradients
- Parameter α sweep (0–1, 11 steps) implemented to explore trade-off solutions between compliance and buckling
- Comparison with OptiStruct on a compliance-only case: compliance +6% (402 vs. 380.5 N·mm), main load paths consistent. (Note: different elements formulation)

Next

- Obtaining a-posteriori Pareto optimality
- Linearized buckling analysis (LBA) validation against commercial software
- Full development of 3D version with the addition of 3D printing constraints (prototype already available on GitHub).
- Further code reviewing and extended tests.

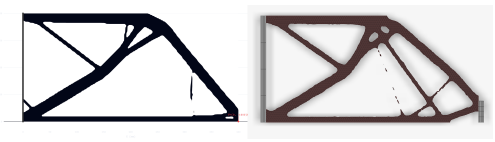


Figure 5: Comparison with OptiStruct of compliance only case

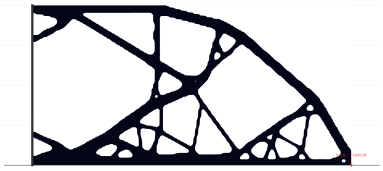


Figure 6: Same cantilever, $\alpha = 0.4$: +21% compliance, BLF × 20 vs. $\alpha = 0$.

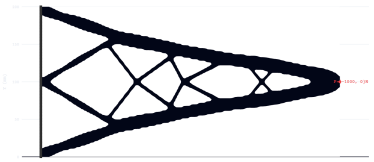


Figure 7: Compressed column, clamped left wall, $V_f = 25\%$, $\alpha = 1$. Qualitatively consistent with Ferrari et al. (2021)